aether that Hendrik Lorentz, Henri Poincaré, Joseph Larmor, and others wrote between 1896 and 1905.

In their articles, Lorentz, Poincaré, and Larmor were groping toward the same revision of our notions of space and time as Einstein, but they were groping through a fog of misconceptions foisted on them by Newtonian physics. Einstein, by contrast, was able to cast off the Newtonian misconceptions. His conviction that the Universe loves simplicity and beauty, and his willingness to be guided by this conviction, even if it meant destroying the foundations of Newtonian physics, led him, with a clarity of thought that others could not match, to his new description of space and time.

The principle of relativity will play an important role later in this book. For this reason I shall devote a few pages to a deeper explanation of it.

A deeper explanation requires the concept of a *reference frame.* A reference frame is a laboratory that contains all the measuring apparatus one might need for whatever measurements one wishes to make. The laboratory and all its apparatus must move through the Universe together; they must all undergo the same motion. In fact, the motion of the reference frame is really the central issue. When a physicist speaks of "different reference frames," the emphasis is on different states of motion and not on different measuring apparatuses in the two laboratories.

A reference frame's laboratory and its apparatus need not be real. They perfectly well can be imaginary constructs, existing only in the mind of the physicist who wants to ask some question such as, "If I were in a spacecraft floating through the asteroid belt, and I were to measure the size of some specific asteroid, what would the answer be?" Such physicists imagine themselves as having a reference frame (laboratory) attached to their spacecraft and as using that frame's apparatus to make the measurement.

Einstein expressed his principle of relativity not in terms of arbitrary reference frames, but in terms of rather special ones: frames (laboratories) that move freely under their own inertia, neither pushed nor pulled by any forces, and that therefore continue always onward in the same state of uniform motion as they began. Such frames Einstein called *inertial* because their motion is governed solely by their own inertia.

A reference frame attached to a firing rocket (a laboratory inside the rocket) is *not* inertial, because its motion is affected by the rocket's thrust as